

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title: Wholesale Mart – B2B Wholesale Order Management System

Skill Task: DBMS

Week: 2

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	INT	Primary Key
Category_name	VARCHAR(40)	UNIQUE
Description	VARCHAR(40)	—

1.2 Product Table

Attribute	Data Type	Constraint
Product_ID	INT	Primary Key
Product_Name	VARCHAR(50)	NOT NULL
Category_ID	INT	Foreign Key
Price	DECIMAL(10,2)	—
MOQ	INT	NOT NULL
Listing_Type	VARCHAR(20)	—
Key_Features	VARCHAR(100)	—
Units_Sold	INT	—
Stock	INT	—

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M) — one category contains many products

3. Table Creation

3.1 Create Category Table

```

1  create table Category(
2      2  Category_ID int primary key,
3      3  Category_name varchar(40) unique,
4      4  Description varchar(40)
5      5  );
6
7  Table created.
8

```

The Category table is created with Category_ID as the primary key and Category_name as a unique field.

3.2 Create Product Table

```

CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50) NOT NULL,
    Category_ID INT,
    Price DECIMAL(10,2),
    MOQ INT NOT NULL,
    Listing_Type VARCHAR(20),
    Key_Features VARCHAR(100),
    Units_Sold INT,
    Stock INT,
    FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)
);
Table created.

```

The Product table is created with Product_ID as the primary key and Category_ID as a foreign key referencing the Category table, establishing a one-to-many relationship.

4. Data Insertion

4.1 Category Records

Ten category records are inserted into the Category table, followed by a commit to save the transaction.

```

INSERT INTO Category VALUES (1, 'Electronics', 'Electronic devices and accessories');
1 row created.
INSERT INTO Category VALUES (2, 'Clothing', 'Men and women clothing products');
1 row created.
INSERT INTO Category VALUES (3, 'Books', 'Books and educational materials');
1 row created.
INSERT INTO Category VALUES (4, 'Home Appliances', 'Appliances used at home');
1 row created.
INSERT INTO Category VALUES (5, 'Sports', 'Sports equipment and accessories');
1 row created.
INSERT INTO Category VALUES (6, 'Beauty', 'Beauty and personal care products');
1 row created.
INSERT INTO Category VALUES (7, 'Groceries', 'Daily food and grocery products');
1 row created.
INSERT INTO Category VALUES (8, 'Furniture', 'Home and office furniture');
1 row created.
INSERT INTO Category VALUES (9, 'Toys', 'Toys and games for children');
1 row created.
INSERT INTO Category VALUES (10, 'Footwear', 'Shoes, sandals and other footwear');
1 row created.

commit;

```

4.2 Product Records

Ten product records are inserted into the Product table, each linked to a category via Category_ID, followed by a commit.

```

insert into Product VALUES (101, 'JBL Earbuds', 1, 2499.00, 10, 'Wholesale', 'Bluetooth
wireless earbuds', 250, 50);
1 row created.
insert into Product VALUES (102, 'Bluetooth Speaker', 1, 3999.00, 5, 'Wholesale', 'Portable
wireless speaker', 180, 30);
1 row created.
insert into Product VALUES (103, 'Wireless Headphones', 1, 5999.00, 5, 'Wholesale', 'Noise
cancelling headphones', 120, 20);
1 row created.
insert into Product VALUES (104, 'Cotton Shirt', 2, 999.00, 10, 'Wholesale', 'Comfortable
cotton shirt', 150, 40);
1 row created.
insert into Product VALUES (105, 'Denim Jeans', 2, 1499.00, 10, 'Wholesale', 'Blue Denim
|jeans', 180, 25);
1 row created.
insert into Product VALUES (106, 'Python Programming Book', 3, 699.00, 5, 'Wholesale', 'Python
programming guide', 80, 15);
1 row created.
insert into Product VALUES (107, 'DBMS Fundamentals', 3, 799.00, 5, 'Wholesale', 'Database
management book', 70, 20);
1 row created.
insert into Product VALUES (108, 'Mixer Grinder', 4, 3499.00, 5, 'Wholesale',
'Kitchen mixer grinder', 60, 10);
1 row created.
insert into Product VALUES (109, 'Electric Kettle', 4, 1999.00, 5, 'Wholesale', 'Stainless
steel electric kettle', 90, 18);
1 row created.
insert into Product VALUES (110, 'Football', 5, 899.00, 10, 'Wholesale', 'Professional
Football', 130, 35);
1 row created.

commit;

```

5. Data Retrieval

5.1 Display Category Records

```
SELECT * FROM Category;
CATEGORY_ID CATEGORY_NAME
-----
DESCRIPTION
▼ -----
| | | 1 Electronics
▼ Electronic devices and accessories
| | | 2 Clothing
▼ Men and women clothing products
| | | 3 Books
Books and educational materials

CATEGORY_ID CATEGORY_NAME
-----
DESCRIPTION
▼ -----
| | | 4 Home Appliances
▼ Appliances used at home
| | | 5 Sports
▼ Sports equipment and accessories
| | | 6 Beauty
Beauty and personal care products

CATEGORY_ID CATEGORY_NAME
-----
DESCRIPTION
▼ -----
```

```
DESCRIPTION
-----
| | | 7 Groceries
Daily food and grocery products
| | | 8 Furniture
Home and office furniture
| | | 9 Toys
Toys and games for children

CATEGORY_ID CATEGORY_NAME
-----
DESCRIPTION
-----
| | | 10 Footwear
Shoes, sandals and other footwear

10 rows selected.
```

All category records are retrieved to verify successful insertion.

5.2 Display Product Records

All product records are retrieved to verify successful insertion.

```
SELECT * FROM Product;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
------------	--------------	-------------

PRICE	MOQ	LISTING_TYPE
-------	-----	--------------

KEY_FEATURES

UNITS_SOLD	STOCK
------------	-------

101	JBL Earbuds	1
2499	10 Wholesale	
Bluetooth wireless	earbuds	
250	50	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
------------	--------------	-------------

PRICE	MOQ	LISTING_TYPE
-------	-----	--------------

KEY_FEATURES

UNITS_SOLD	STOCK
------------	-------

102	Bluetooth Speaker	1
3999	5 Wholesale	
Portable wireless	speaker	
180	30	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
103	Wireless Headphones	1
5999	5 Wholesale	
Noise cancelling headphones		
120	20	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
104	Cotton Shirt	2
999	10 Wholesale	
Comfortable cotton shirt		
150	40	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
105	Denim Jeans	2
1499	10 Wholesale	
Blue denim jeans		
100	25	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
106	Python Programming Book	3
699	5 Wholesale	
Python programming guide		
80	15	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
107	DBMS Fundamentals	3
799	5 Wholesale	
Database management book		
70	20	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
108	Mixer Grinder	4
3499	5 Wholesale	
Kitchen mixer grinder		
60	10	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
109	Electric Kettle	4
1999	5 Wholesale	
Stainless steel electric kettle		
90	18	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
110	Football	5
899	10 Wholesale	
Professional football		
130	35	

10 rows selected.

6. Product Update

Update the price of JBL Earbuds (Product_ID 101).

```
UPDATE Product
SET Price = 2299.00
WHERE Product_ID = 101;
1 row updated.
```

Verify the updated price by selecting the record.

```
SELECT * FROM Product
WHERE Product_ID = 101;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
101	JBL Earbuds	1
2299	10 Wholesale	
Bluetooth wireless earbuds		
250	50	

Update the stock quantity of the same product and commit the changes.

```
UPDATE Product
SET Stock = 60
WHERE Product_ID = 101;
1 row updated.

commit;
```

7. Product Deletion

Delete the product Football (Product_ID 110) from the Product table.

```
DELETE FROM Product
WHERE Product_ID = 110;
1 row deleted.
```

Verify the deletion by displaying the remaining product records.

```
SELECT * FROM Product;
PRODUCT_ID PRODUCT_NAME                                CATEGORY_ID
-----
PRICE      MOQ LISTING_TYPE
-----
KEY_FEATURES
-----
UNITS_SOLD  STOCK
-----
101 JBL Earbuds                                1
2299      10 Wholesale
Bluetooth wireless earbuds
250      60

PRODUCT_ID PRODUCT_NAME                                CATEGORY_ID
-----
PRICE      MOQ LISTING_TYPE
-----
KEY_FEATURES
-----
UNITS_SOLD  STOCK
-----
102 Bluetooth Speaker                            1
3999      5 Wholesale
Portable wireless speaker
180      30
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
103	Wireless Headphones	1
5999	5 Wholesale	
Noise cancelling headphones		
120	20	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
104	Cotton Shirt	2
999	10 Wholesale	
Comfortable cotton shirt		
150	40	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
105	Denim Jeans	2
1499	10 Wholesale	
Blue denim jeans		
100	25	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
106	Python Programming Book	3
699	5 Wholesale	
Python programming guide		
80	15	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
107	DBMS Fundamentals	3
799	5 Wholesale	
Database management book		
70	20	
PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
108	Mixer Grinder	4
3499	5 Wholesale	
Kitchen mixer grinder		
60	10	

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID
PRICE	MOQ LISTING_TYPE	
KEY_FEATURES		
UNITS_SOLD	STOCK	
109	Electric Kettle	4
1999	5 Wholesale	
Stainless steel electric kettle		
90	18	

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the category name along with the products belonging to each category, using an inner JOIN between Category and Product.

Conclusion

The Category and Product tables were successfully designed, created, and populated with sample data. CRUD operations (Insert, Select, Update, Delete) were performed and verified, and a JOIN query was used to generate a category-wise product report, demonstrating a working one-to-many relationship between Category and Product.